

Communication

Efficient Antenna Modeling and Optimization Using Multifidelity Stacked Neural Network

Ju Tan¹, Yu Shao², Jiliang Zhang³, and Jie Zhang⁴

Abstract—In this work, a multifidelity stacked neural network (MFSNN) is proposed to construct surrogate model for antenna modeling and optimization. The stacked neural network consists of a low-fidelity (LF) network, a linear high-fidelity (HF) network, and a nonlinear HF network. By learning the prior from sufficient computationally cheap LF data, the MFSNN has significantly reduced the requirement of computationally expensive HF data. The correlation between LF and HF models can be learned adaptively and accurately by decomposing the correlation into linear component and nonlinear component. The feasibility of the approach is validated by two antenna structures, which shows that the MFSNN-based surrogation model can make predictions for broad ranges of input parameters with satisfactory accuracy. Then, the surrogate model is directly applied in the particle swarm optimization (PSO) framework to replace the full-wave simulation and accelerate antenna optimization procedure.

Index Terms—Antenna modeling, antenna optimization, multifidelity, stacked neural network, surrogate model.

I. INTRODUCTION

The computational electromagnetics (EMs)-based simulation is an accurate and reliable way to evaluate antenna. However, with the rapid development of modern wireless communications, the structure of antennas is getting more complex, which leads to more requirements and constraints for design and optimization. In this premise, the full-wave EM simulation solver is computationally intensive once the more complex antenna structures and increasing number of design variables are considered. Conventional antenna optimization routine posts a challenge, since a large number of evaluations of antennas are required to search for the optimal antenna design, especially global optimization using meta-heuristics algorithm (e.g., particle

swarm optimization (PSO) [1] and genetic algorithms (GAs) [2]) is extremely computationally expensive.

Surrogate modeling is an effective method to address the above issue, in which a computationally cheap yet accurate surrogate is trained to replace the full-wave EM simulation, so that the candidates in the global optimization can be evaluated fast [3]. A surrogate model can be constructed by a supervised machine learning method, such as artificial neural network (ANN) and Gaussian process regression (GPR) [4]. ANN is a powerful tool for data-driven modeling and has been introduced into antenna design and optimization for substituting the computationally intensive EM simulation many years ago [5], [6]. An ANN synthesis model [7] and an ANN-based multiparameter model [8] were developed to facilitate the procedure of antenna modeling, and it was shown that the results predicted by ANN agree well with the EM simulation. Recently, a consensus deep neural network [9] was developed to reduce the uncertainty of a single ANN and to accelerate the antenna optimization. In [10], an efficient knowledge-based ANN was proposed for the design of circularly polarized lens antenna with multiobjectives. A fully automated surrogate model was proposed in [11], in which the architecture of ANN can be automatically determined using Bayesian optimization. In [12] and [13], a domain-constrained deep neural network model was proposed for accurate modeling of antennas and microwave passives with low-computational cost. However, a large amount of training data are required to ensure the accuracy of the ANN surrogate model. Inadequate training data can lead to inaccurate surrogate and overfitting issues. Since the training data are usually generated by high-fidelity (HF) EM simulation, the time cost of obtaining adequate training data is high. In this case, the GPR becomes popular in antenna design, because it can give accurate predictions using less training set and provide confidence interval for the predictions [14]. However, the time cost of obtaining training data using EM simulation is still high.

A multifidelity modeling method can significantly reduce the cost of gathering initial training data for surrogate model construction by leveraging both low-fidelity (LF) and HF data [15]. In EM simulation, the fidelity is generally defined according to the mesh size: HF models are generated by the highly accurate and computationally expensive simulations with fine discretization, while the LF models obtained by simulations with coarse mesh size are less accurate, but their evaluations are very fast [16]. By exploiting the correlation between LF and HF models, sufficient LF data can be utilized along with a small amount of HF data to accurately build up a surrogate model. The combination of GPR and multifidelity modeling is popular in antenna design. In [17], a cokriging model was constructed using sufficient LF data and a small quantity of HF data. In [18], a two-stage framework was proposed for efficient antenna modeling using GPR, which could learn the mapping between models of different fidelities and reduce the number of HF simulations substantially. A multistage modeling method was proposed in [19] to learn the mappings between the EM models with variable fidelity. In [20], a surrogate-assisted quasi-Newton-enhanced global optimization method was presented, and the multifidelity models were used to reduce the cost

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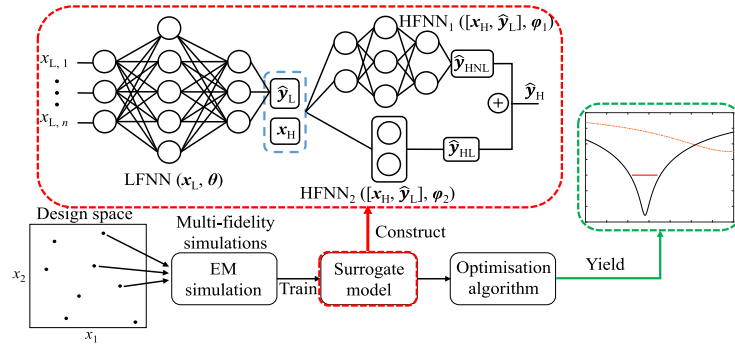


Fig. 1. Schematic of MFSNN for antenna modeling and optimization.

of individual simulation. Pietrenko-Dabrowska et al. [21] and Koziel and Pietrenko-Dabrowska [22] proposed the cokriging surrogate model employing domain confinement approach combined with variable-fidelity EM simulations. Global optimization schemes were presented in [23] and [24], in which the computation costs were reduced by the incorporation of variable-fidelity EM simulations.

Compared to GPR, ANN can be flexibly interfered with (e.g., add and drop neurons or layers, train from checkpoints) and transferred. By applying techniques, such as regularization and dropout, ANN can obtain comparable efficiency and accuracy as GPR. The combination of a multifidelity approach and the deep neural network has been applied to fluid mechanics [25] and partial differential equation (PDE) solving [26]. The results indicate that the multifidelity-based neural network has excellent expressive capability and can mitigate the overfitting issue, while a small amount of training data are used. However, to the best of our knowledge, multifidelity-based neural network applied in antenna modeling and optimization has not been reported yet. On the other hand, the multifidelity-based GPR framework in current antenna modeling and optimization works is based on a linear autoregressive scheme, which assumes that the correlation between LF and HF data is linear. However, the correlation between different fidelities can go beyond linearity if the ranges of input parameters are large [27]. In this communication, an MFSNN is proposed to construct surrogate model for antenna optimization [28]. The specific contributions of this work include the following.

- 1) A multifidelity-based stacked neural network is introduced in antenna modeling, which significantly reduces the HF training datasets required for ANN surrogate model construction.
- 2) Both linear and nonlinear correlations between LF and HF models are considered in order to adaptively exploit the multifidelity information fusion in antenna modeling, which improves the accuracy of the proposed surrogate model.
- 3) To demonstrate the excellent capability of nonlinear relationship learning and generalization of the proposed ANN-based surrogate, accurate global models for broad ranges of input parameters are constructed, which can be applied in the global optimization framework directly and greatly simplify the optimization process.

II. OVERVIEW OF MULTIFIDELITY NEURAL NETWORK

A surrogate composed of three full-connected independent neural networks derived from [26] is presented in Fig. 1, which are an LF neural network (LFNN), a nonlinear HF neural network (HFNN₁), and a linear HFNN (HFNN₂). The multifidelity modeling method is applied to the neural network construction, which aims to exploit the relationship between LF and HF data [26]. The LF response y_L is defined as follows:

$$y_L = g(x_L) \quad (1)$$

where $g(\cdot)$ is unknown function, which calculates the LF response of input x_L . The LFNN (x_L, θ) is used to approximate $g(\cdot)$ to give predicted LF response $\hat{y}_L$, where θ is the corresponding

hyperparameters. The relationship between LF and HF data could be expressed as follows:

$$y_H = f(x_H, \hat{y}_L) \quad (2)$$

where $f(\cdot)$ is an unknown function that is used to reveal the correlation between different fidelities and y_H is the HF response. HFNN ($x_H, \hat{y}_L, \phi$) is used to approximate $f(\cdot)$ to give a predicted HF responses $\hat{y}_H$, where $\hat{y}_L$ is the predicted LF response by LFNN for the input x_H and ϕ represents the hyperparameters. In antenna design, the input $x(x_L, x_H)$ is the vector of design space variables, and the output response $y(y_L, y_H)$ is the vector of performance metrics of antenna, such as S_{11} (real and imagery parts), axial ratio, and so on. Considered the potential correlation between LF and HF data, $f(\cdot)$ is decomposed into nonlinear component and linear component, which are modeled by HFNN₁ and HFNN₂ with hyperparameters of ϕ_1 and ϕ_2 , respectively, as shown in Fig. 1. HFNN₁ and HFNN₂ are trained parallelly, and the final output $\hat{y}_H$ of the entire network is the sum of the two HFNN's outputs $\hat{y}_{HNL}$ and $\hat{y}_{HL}$; in this way, the correlation between LF and HF data can be learned adaptively.

In the training process, the bias and weights of the neural network are adjusted by minimizing the cost functions and optimizing the hyperparameters. In our stacked ANN framework, each network has individual cost function, and the total cost of the whole network is measured as the sum of individual costs and regularization item cost, which are shown as follows:

$$f_{\text{total}} = f_{\text{LFNN}} + f_{\text{HFNN}} + \lambda \sum_{i=1}^M \sum_{j=1}^{N_i} \omega_{i,j}^2 \quad (3)$$

where

$$f_{\text{LFNN}} = \frac{1}{N_{\text{LF}}} \sum_{n=1}^{N_{\text{LF}}} |\hat{y}_{\text{LF}} - y_L|^2 \quad (4)$$

$$f_{\text{HFNN}} = \frac{1}{N_{\text{HF}}} \sum_{n=1}^{N_{\text{HF}}} |\hat{y}_{\text{HNL}} - y_H|^2 + \frac{1}{N_{\text{HF}}} \sum_{n=1}^{N_{\text{HF}}} |\hat{y}_{\text{HL}} - y_H|^2 \quad (5)$$

$w_{i,j}$ is the weight of the j th neuron in the i th hidden layer of HFNN₁, N_i is the number of neurons in the i th hidden layer, and M is the number of hidden layers in HFNN₁. λ is the L_2 regularization ratio for $w_{i,j}$, which is set to be 0.001 in this work. L_2 regularization is an extensive strategy to prevent the potential overfitting issue, especially when the amount of training data are small. The network is trained using the Adam algorithm together with the Xavier's initialization method, and the limited memory BFGS (L-BFGS) algorithm is employed for searching optimum locally. The training termination criterion is the cost function value less than 10^{-6} .

The method starts with obtaining the training data using EM simulation with different fidelities. In this communication, the commercial EM simulation software CST Microwave Studio is used. The LFNN is first constructed using LF simulated results; then, its prediction on the input of the HF dataset is stacked with HF data input x_H , plus with the HF simulated results, to form a new input and train the

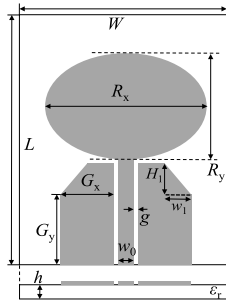


Fig. 2. Structure of the wideband CPW-fed monopole antenna (Antenna 1).

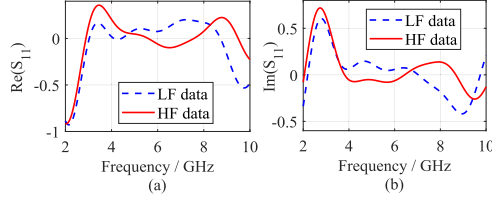


Fig. 3. Response of LF and HF models for Antenna 1. (a) $\text{Re}(S_{11})$ and (b) $\text{Im}(S_{11})$ against frequency. Antenna geometry $\mathbf{x} = [7.31, 9.90, 10.56, 6.39]$.

HFNN. The HF results are obtained by simulation with the default fine mesh density in CST, while the LF results are simulated with the following: 1) coarse mesh (reducing the number of meshes); 2) relaxed steady-state criteria (reducing the simulation duration); 3) simplified excitation port (e.g., discrete port instead of waveguide port); and 4) simplified material (e.g., perfect conductor and dielectric without loss). After training, the whole network is then utilized to approximate the HF data and make a prediction on new design points.

III. SURROGATE MODELING ON ANTENNAS

A. Wideband Monopole Antenna (Antenna 1)

The antenna structure is shown in Fig. 2, in which an elliptical monopole is fed by a 50- Ω coplanar waveguide (CPW) line with chambered ground [29]. The design variable is $\mathbf{x} = [R_x, R_y, G_x, G_y]$, whose lower and upper bounds are [5, 5, 6, 6] mm and [15.5, 15.5, 11, 11] mm, respectively. Other dimensions are fixed (substrate dielectric constant $\epsilon_r = 4.2$, substrate thickness $h = 1.6$ mm, $H_1 = 7$ mm, $w_1 = 3$ mm, $w_0 = 1.6$ mm, and the width W and length L of the dielectric substrate are 31 and 44.3 mm, respectively). The frequency band of interest is 2–10 GHz, and the modeling interests are the real part and imaginary part of the reflection coefficient ($\text{Re}\{S_{11}\}$ and $\text{Im}\{S_{11}\}$) within the frequency band of interest.

The training datasets consist of 200 geometries selected using Latin hypercube sampling (LHS) within the design space, with 50 uniformly sampled frequency points per geometry. The testing data contain 100 new geometries, which are also obtained by LHS, with 121 equally spaced frequency points for each geometry.

The LF and HF antenna models are constructed and simulated using CST on a desktop with Intel Xeon W2135 3.7-GHz CPU and 32-GB RAM. The HF model for Antenna 1 with fine mesh density and high accuracy is constructed using 274 220 mesh cells, and the simulation time is 213 s. In contrast, the LF model is constructed with 12 354 mesh cells, and the simulation time is 10 s. The responses of the HF and LF model for one sample are presented in Fig. 3. It is observed that although the LF model is not as accurate as HF model, they have the similar overall trend.

The training data are simulated as both LF and HF models to construct the surrogate model, while the testing data are only simulated as the HF model for testing the generalization ability of the proposed MFSNN. The surrogate modeling is repeatedly implemented using different proportions of HF data, the proportion is

TABLE I
TRAINING AND TESTING ERRORS¹ OF ANTENNA 1 MODELING WITH DIFFERENT HF DATA PROPORTIONS

HF proportion (training time)	Tr ² [Re ³]	Tr[Im ³]	Te ² [Re]	Te[Im]	Te $ S_{11} $ (dB) ⁴
100% (2.4 h)	0.001%	0.002%	0.11%	0.15%	0.14%
70% (3.7 h)	0.001%	0.001%	0.13%	0.07%	0.15%
60% (3.5 h)	0.002%	0.003%	0.26%	0.17%	0.28%
50% (3.5 h)	0.003%	0.003%	0.39%	0.24%	0.46%
40% (3.3 h)	0.005%	0.004%	0.50%	0.38%	0.89%
30% (3.4 h)	0.009%	0.007%	0.65%	0.42%	0.97%
20% (3 h)	0.013%	0.011%	1.13%	1.49%	1.67%
10% (2.8 h)	0.021%	0.016%	2.07%	2.08%	2.42%

¹ Normalised mean square error in percentage: $\frac{\sum_{i=1}^{N_{\text{HF}}} |\hat{y} - y_{\text{HF}}|^2}{\sum_{i=1}^{N_{\text{HF}}} |y_{\text{HF}}|^2}$

² Tr stands for training, Te stands for testing.

³ Re and Im stand for real part and imaginary part of S_{11} .

⁴ Errors calculated using $|S_{11}|$ in dB.

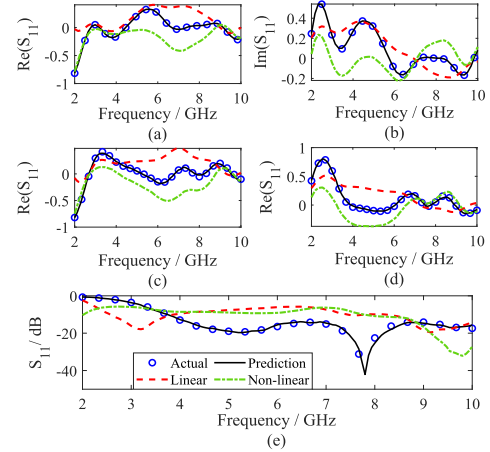


Fig. 4. Antenna 1 modeling results using 10% HF data. (a) Training result of $\text{Re}(S_{11})$. (b) Training result of $\text{Im}(S_{11})$. (c) Testing result of $\text{Re}(S_{11})$. (d) Testing result of $\text{Im}(S_{11})$. (e) Testing result of $|S_{11}|$ (dB). Antenna geometry: training $\mathbf{x} = [12.44, 8.49, 6.60, 10.50]$ and testing $\mathbf{x}^* = [14.01, 8.38, 8.63, 6.68]$.

defined as $\frac{n_{\text{HF}}}{n} \times 100\%$, n is the number of total geometries using in the training data, which is 200 in this example, and n_{HF} is the number of geometries that are simulated using HF simulation. The percentage of HF data in training dataset varies from 70% to 10%. For example, if the percentage of HF data is 10%, we use 200 LF datasets to train LFNN, and 20 HF datasets to train HFNN₁ and HFNN₂.

LFNN is constructed with nine hidden layers with (30, 60, 100, 120, 180, 120, 100, 60, 30) neurons in each layer. HFNN₁ has nine hidden layers with (30, 60, 100, 120, 180, 120, 100, 60, 30) neurons in each layer, while HFNN₂ has only one linear layer. All the hidden neurons in LFNN and HFNN₁ use Leaky ReLU activation function, and the linear transfer unit is used for output layers for all networks. The initial learning rate is set to be 0.001, which is scheduled to decay if the cost function does not change for 100 epochs.

The training and testing errors are listed in Table I, where the 100% proportion of HF data are set as a reference, which means all the 200 training datasets are generated by HF simulation and are used to train a conventional ANN. The training time of the surrogate model is also shown in Table I. The training and testing results of the surrogate model using 10% HF data are shown in Fig. 4, in which the individual output responses of HFNN₁ and HFNN₂, as well as the prediction result of the whole MFSNN (summation of the responses of HFNN₁ and HFNN₂), are compared with the actual response. Since the simulation time of the LF model is only about one-twentieth of the HF model, we use large amount of LF data to train LFNN to predict the variation trend of the S_{11} . After that, a small amount of HF data are used to train HFNN to compensate the discrepancy between LF and HF models. Table I and Fig. 4 show

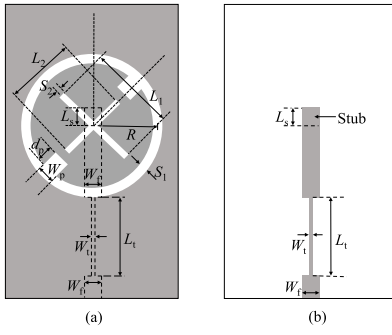


Fig. 5. Structure of the dual-frequency slot antenna (Antenna 2). (a) Front view. (b) Back view.

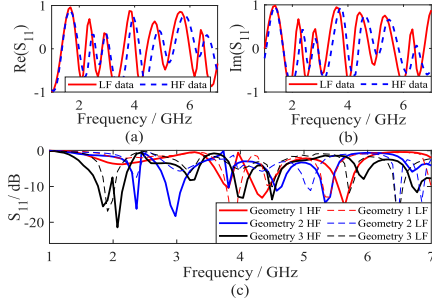


Fig. 6. Response of LF and HF model for Antenna 2. (a) $\text{Re}(S_{11})$ and (b) $\text{Im}(S_{11})$ against frequency, antenna geometry $\mathbf{x} = [22.34, 11.9, 12.17, 5.22, 5.63, -0.62, 15.36, 1.22]$. (c) Variation of $|S_{11}|$ for three geometries distributed throughout the design space, antenna geometries $\mathbf{x}_1 = [24.94, 22.82, 13.47, 1.09, 1.89, -1.77, 11.8, 2.46]$; $\mathbf{x}_2 = [26.94, 23.70, 14.47, 1.29, 3.54, 3.88, 10.53, 0.42]$; $\mathbf{x}_3 = [24.47, 22.35, 17.02, 5.10, 2.14, 4.29, 22.62, 1.03]$.

that even if the amount of HF data are only 10% of LF data, the prediction error on testing dataset is about 2%. The proposed MFSNN has greatly reduced the demand for HF training datasets, which can be served as a good global surrogate model, since it exhibits high accuracy in a wide range of design variables.

B. Dual-Frequency Dual-Sense Circularly Polarized Slot Antenna (Antenna 2)

Fig. 5 presents a dual-frequency dual-sense circularly polarized slot antenna [30] composed by an annular slot and a cross slot fed by microstrip line. The considered design vector is $\mathbf{x} = [L_1, L_2, R, d_p, W_p, L_s, L_t, W_t]$, where L_1 and L_2 are the lengths of the two rectangular slots forming the cross slot, R is the radius of the annular slot, d_p and W_p are the depth and width of the pair of notches on the annular slot, L_s is the length of the stub of the microstrip line, and L_t and W_t are the length and width of the impedance transformer. The lower bound and upper bound of the variation range of design vector are $[22, 18, 12, 1, 1, -2, 10.5, 0.3]$ mm and $[34, 32, 18, 6, 6, 5, 22.8, 2.8]$ mm, respectively. Other parameters are substrate thickness $h = 1.45$ mm, dielectric constant of substrate $\epsilon_r = 4.2$, and width of the feeding line $W_f = 2.96$ mm. The modeling targets are the real part and imaginary part of S_{11} over 1–7 GHz, and the axial ratio within 2–3 and 5–6.5 GHz.

For training purposes, 700 geometries are sampled within the design space using LHS with 50 uniformly sampled frequency points per geometry. The testing dataset is composed of 100 new geometries obtained using LHS, with 121 equally spaced frequency points per geometry.

The HF model for Antenna 2 is constructed with 370 818 mesh cells in CST, and its simulation costs 324 s, while the LF model is constructed in 16 770 mesh cells whose simulation time is 12 s. We can see that the simulation time of LF model is 27 times smaller than that of the HF model. The antenna responses of different

TABLE II
TRAINING AND TESTING ERRORS¹ OF ANTENNA 2 MODELING WITH DIFFERENT HF DATA PROPORTIONS

HF proportion (training time)	$\text{Tr}^2[\text{Re}^3]\text{Tr}[\text{Im}^3]$	$\text{Te}^2[\text{Re}]\text{Te}[\text{Im}]$	$\text{Te} S_{11} $ (dB) ⁴	$\text{Tr}(\text{AR}^5)\text{Te}(\text{AR})$			
100% (6.2 h)	0.17%	0.19%	4.32%	3.68%	3.84%	0.01%	0.71%
70% (8.1 h)	0.23%	0.21%	6.13%	5.37%	4.03%	0.07%	1.53%
60% (7.4 h)	0.34%	0.23%	6.93%	7.17%	5.95%	0.15%	1.96%
50% (7.3 h)	0.37%	0.28%	6.56%	6.42%	6.18%	0.24%	3.29%
40% (6.8 h)	0.38%	0.36%	7.89%	7.80%	7.90%	0.29%	5.26%
30% (6.5 h)	0.32%	0.39%	8.84%	8.26%	9.95%	0.41%	5.38%

¹ Normalised mean square error in percentage: $\frac{\sum_{1}^{N_{\text{HF}}} |\hat{y} - y_{\text{HF}}|^2}{\sum_{1}^{N_{\text{HF}}} |y_{\text{HF}}|^2}$

² Tr stands for training, Te stands for testing.

³ Re and Im stand for real part and imaginary part of S_{11} .

⁴ Errors calculated using $|S_{11}|$ in dB.

⁵ AR stands for axial ratio.

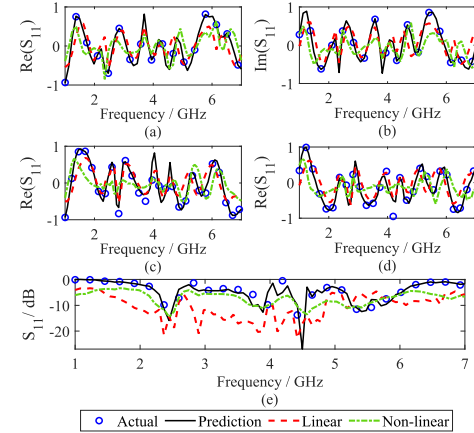


Fig. 7. Antenna 2 modeling results using 30% HF data. (a) Training result of $\text{Re}(S_{11})$. (b) Training result of $\text{Im}(S_{11})$. (c) Testing result of $\text{Re}(S_{11})$. (d) Testing result of $\text{Im}(S_{11})$. (e) Testing result of $|S_{11}|$ (dB). Antenna geometry: training $\mathbf{x} = [29.03, 22.62, 4.17, 13.91, 17.22, 5.33, 0.8, 4.91]$ and testing $\mathbf{x}^* = [25.51, 22.86, -0.75, 10.70, 16.35, 3.22, 2.36, 3.92]$.

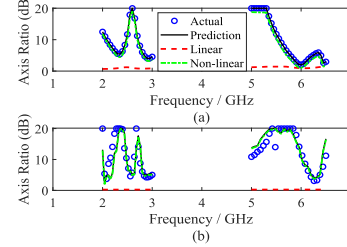


Fig. 8. Antenna 2 axial ratio modeling results using 30% HF data. (a) Training result. (b) Testing result. Antenna geometry: training $\mathbf{x} = [26.40, 21.28, 3.83, 23.73, 14.20, 2.87, 0.41, 2.56]$ and testing $\mathbf{x}^* = [25.58, 14.44, 2.24, 13.39, 13.79, 2.31, 1.63, 5.57]$.

fidelties are shown in Fig. 6(a) and (b). Fig. 6(c) shows $|S_{11}|$ for three randomly selected geometries, which indicate substantial variations of S_{11} in a broad design space. This work aims to construct a global surrogate model to predict the variation of S_{11} accurately throughout the broad design space.

The modeling process is similar to Antenna 1, and the proportion of HF data in training dataset varies from 70% to 30%. LFNN is constructed with 12 hidden layers in with (30, 60, 100, 120, 240, 300, 240, 180, 120, 100, 60, 30) neurons in each hidden layer. HFNN₁ has ten hidden layers with (30, 60, 100, 120, 240, 180, 120, 100, 60, 30) neurons in each hidden layer, while HFNN₂ has only one linear layer. All the hidden neurons in LFNN and HFNN₁ use Leaky ReLU activation function, and the linear transfer unit is used for output layers for all networks. The initial learning rate is set to be 0.002, which is also scheduled to decay if the cost function keeps constant for 100 epochs.

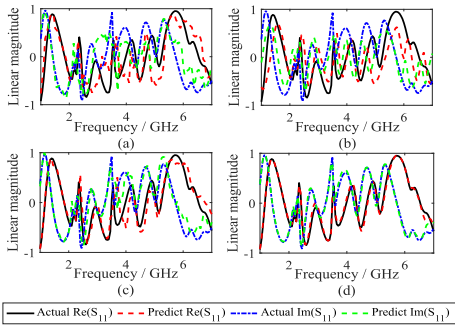


Fig. 9. Testing results for Antenna 2 modeling with 30% HF data using (a) conventional ANN, (b) LFNN + HFNN₂, (c) LFNN + HFNN₁, and (d) LFNN + HFNN₁ + HFNN₂.

The corresponding training and testing errors as well as training time are listed in Table II. The training and testing S_{11} and axial ratio results of the surrogate model on Antenna 2 using 30% HF data are shown in Figs. 7 and 8, respectively. In this example, the dimension of the design space is high, and the variation range of each design variable is large. Table II, and Figs. 7 and 8 indicate that when the percentage of HF data is 30%, the testing errors on new geometries are less than 10%. So, the proposed MFSNN framework can construct an accurate global surrogate model for complex antenna structure with just a little amount of HF data.

To investigate the effect of decomposition in HF data approximation, the modeling results using 30% HF data (210 geometries are evaluated as the HF model) by different manners are presented in Fig. 9. Fig. 9(a) shows the prediction results of a single conventional ANN trained by only 210 HF datasets. Because the training data are insufficient, the conventional ANN cannot extract the features from the training data and the generalization ability is limited, thus giving inaccurate prediction (prediction errors are 76.52% and 67.15% for real and imaginary parts of S_{11} , respectively, for this specific testing sample). For the result in Fig. 9(b), both LFNN and HFNN are employed in modeling; however, only linear network (HFNN₂) is used in HFNN. The accuracy for this situation is poor (with the errors of 47.29% and 41.94% for real and imaginary parts, respectively), which indicates that the correlation between LF and HF models goes beyond linearity, especially for higher frequency band, since coarse mesh leads to large error at higher frequency where the discrepancy between LF and HF models is complex. Fig. 9(c) gives the modeling results of LFNN connected with a nonlinear network (HFNN₁). Compared to the results in Fig. 9(b), the surrogate model in this case obtains higher accuracy except deteriorating in a few details (with errors of 21.70% for real part and 17.47% for imaginary part), and it appears this neural network has learned most of the features of the data. The modeling results of the proposed MFSNN are shown in Fig. 9(d), in which the prediction results agree well with the actual HF results (with the errors of 8.37% and 8.13% for real and imaginary parts, respectively, consistent with the testing errors reported in Table II). Compared to Fig. 9(a), MFSNN model yields accurate prediction with small amount of HF data, since it learns prior knowledge from sufficient LF data; compared to Fig. 9(b) and (c), the MFSNN gains the highest prediction accuracy, since it decomposes the correlation between LF and HF models into linear and nonlinear components, so that the correlation can be learned adaptively.

IV. APPLICATION ON ANTENNA OPTIMIZATIONS

In this section, the aforementioned MFSNN surrogate model is applied on antenna optimization to validate the accuracy and efficiency of the proposed modeling method. The two antennas discussed in Section III are employed to demonstrate the optimization process. The objective is to minimize the maximum $|S_{11}|$ within

TABLE III
PSO PARAMETERS FOR ANTENNA OPTIMIZATION

Parameters	Antenna 1	Antenna 2
Swarm size	30	100
Velocity boundary	± 0.8	± 0.5
Learning factor c_1	1.4	1.8
Learning factor c_2	1.5	2
Inertial weight W	[0.2, 1]	[0.2, 1]

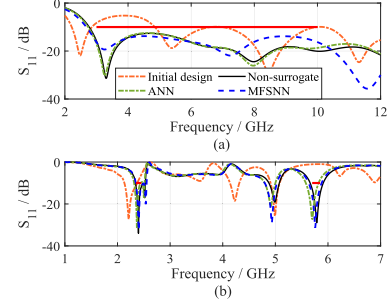


Fig. 10. Antenna optimization results of the three methods along with the responses of initial design for (a) Antenna 1 and (b) Antenna 2. The design objectives are drawn with red solid line.

the frequency band of interest (3.1–10.6 GHz for Antenna 1, and 2.35–2.45 and 5.75–5.85 GHz for Antenna 2). The PSO algorithm is utilized to search the optimum candidate in the design space. For the purpose of comparison, the optimization is implemented based on three methods. In method 1, no surrogate model is used, and each candidate in PSO is evaluated by HF full-wave simulation. In method 2, the surrogate model is constructed by the conventional ANN, which is trained by sufficient HF data. In method 3, the surrogate model is constructed by the proposed MFSNN, which is trained by sufficient LF data along with a small amount of HF data. The proportion of HF data is 10% for Antenna 1 and 30% for Antenna 2. In methods 2 and 3, the candidates in PSO are evaluated by the corresponding surrogate model. For each antenna, the PSO parameters are the same in the three implementation methods, which are listed in Table III.

The antenna optimization results are shown in Fig. 10, in which the HF full-wave responses of initial design and optimized designs for the three methods are provided. It can be seen that all the three methods can obtain the optimized design, which fulfills the objective. Although the proposed MFSNN introduces some modeling errors due to the limited amount of HF training data, it is accurate enough to surrogate the full-wave simulation in a wide design space for optimization application.

The comparisons of the cost of the three methods are shown in Table IV, in which the maximum $|S_{11}|$ in the frequency band of interest, the number of antenna evaluations in PSO, the number of full-wave EM simulations, and the amount of total time cost are given. The maximum $|S_{11}|$ is below -10 dB for all the three methods of each antenna after optimization. In the nonsurrogate method, every candidate in each iteration of PSO has to call HF full-wave EM simulation to evaluate the antenna performance, so the computational cost is very high. In contrast, the cost of candidate antenna evaluation in surrogate-based optimization can be neglected, since it is just a fast forward pass of neural network in our model. However, the construction of conventional ANN requires a large number of HF training data to ensure its prediction accuracy. Although the ANN method eliminates the full-wave simulation in optimization stage, it heavily relies on HF EM simulation in construction stage, so its full-wave EM simulation cost is comparable or even larger (for Antenna 1) than the non-surrogate method. The proposed MFSNN has significantly reduced the requirement of HF EM simulations in model construction by learning the prior from LF data and exploiting the correlation between LF and HF models, and it can be applied in global optimization

TABLE IV
COMPARISONS OF THE THREE OPTIMIZATION METHODS

Antenna	Method	$\max S_{11} ^2$ (dB)	Number of antenna evaluation ³	Number of EM simulations ⁴	Total time cost ⁵ (h)
1	NS ¹	-12.64	150	150 (H)	8.88
	ANN	-12.50	120	200 (H)	14.23
	MFSNN	-13.80	210	20 (H) + 180 (L)	4.48
2	NS	-11.94	1000	1000(H)	90
	ANN	-10.16	900	700 (H)	69.2
	MFSNN	-11.39	1100	210 (H) + 490 (L)	27.03

¹ NS stands for non-surrogate method.

² maximum value of $|S_{11}|$ at frequency band of interest: 3.1 to 10.6 GHz for Antenna 1, and 2.35 to 2.45 GHz and 5.75 to 5.85 GHz for Antenna 2, optimisation objective is $\max|S_{11}| \leq -10$ dB.

³ Total number of antenna evaluations during the optimisation process.

⁴ It counts the total required number of EM simulations in surrogate model construction stage and optimisation stage; where (H) refers to EM simulation of HF models, (L) is for LF models.

⁵ Including time cost of EM simulations in surrogate model construction stage and optimisation stage, as well as surrogate model training.

framework directly to replace the computationally expensive full-wave simulation and accelerate the optimization procedure.

V. CONCLUSION

A stacked neural network based on multifidelity framework is introduced for antenna modeling and optimization, which can construct an accurate surrogate model in a wide design space with just a small amount of HF training data. The correlation between LF and HF models can be learned adaptively and accurately by decomposing the correlation into linear component and nonlinear component. Two antenna examples are used to demonstrate the model construction, and the modeling errors are discussed in different cases. Furthermore, the application on antenna optimization is presented, and the performances of different optimization methods are compared. Last but not least, the MFSNN-based surrogate model presented in this communication can be efficiently applied in but not limited to optimization, robust design, and sensitivity analysis. More application scenarios are future research topics.

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